

Input Output in java :

In order to perform input and output operation, Java software people has provided a predefined package called java.io.
By using this package we can read the data from keyword (DataInputStream, BufferedRedaer), can create a file, can perform read and write operation to the file.

What is the need of File Handling :

File handling is nothing but creating a file and performing read and write operation to the file.
As we know to store a value in the program we need to take the support of variable/Field.

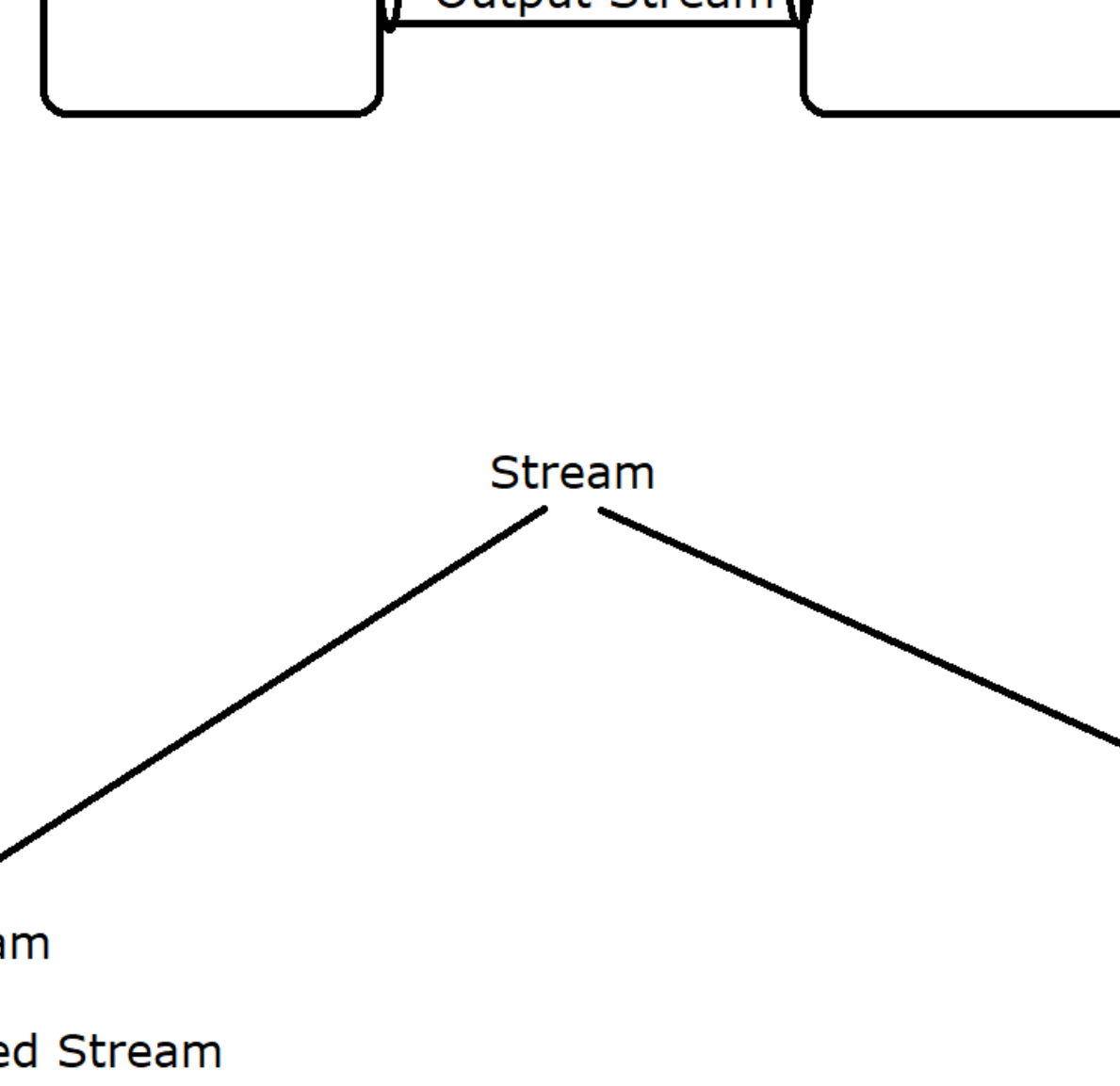
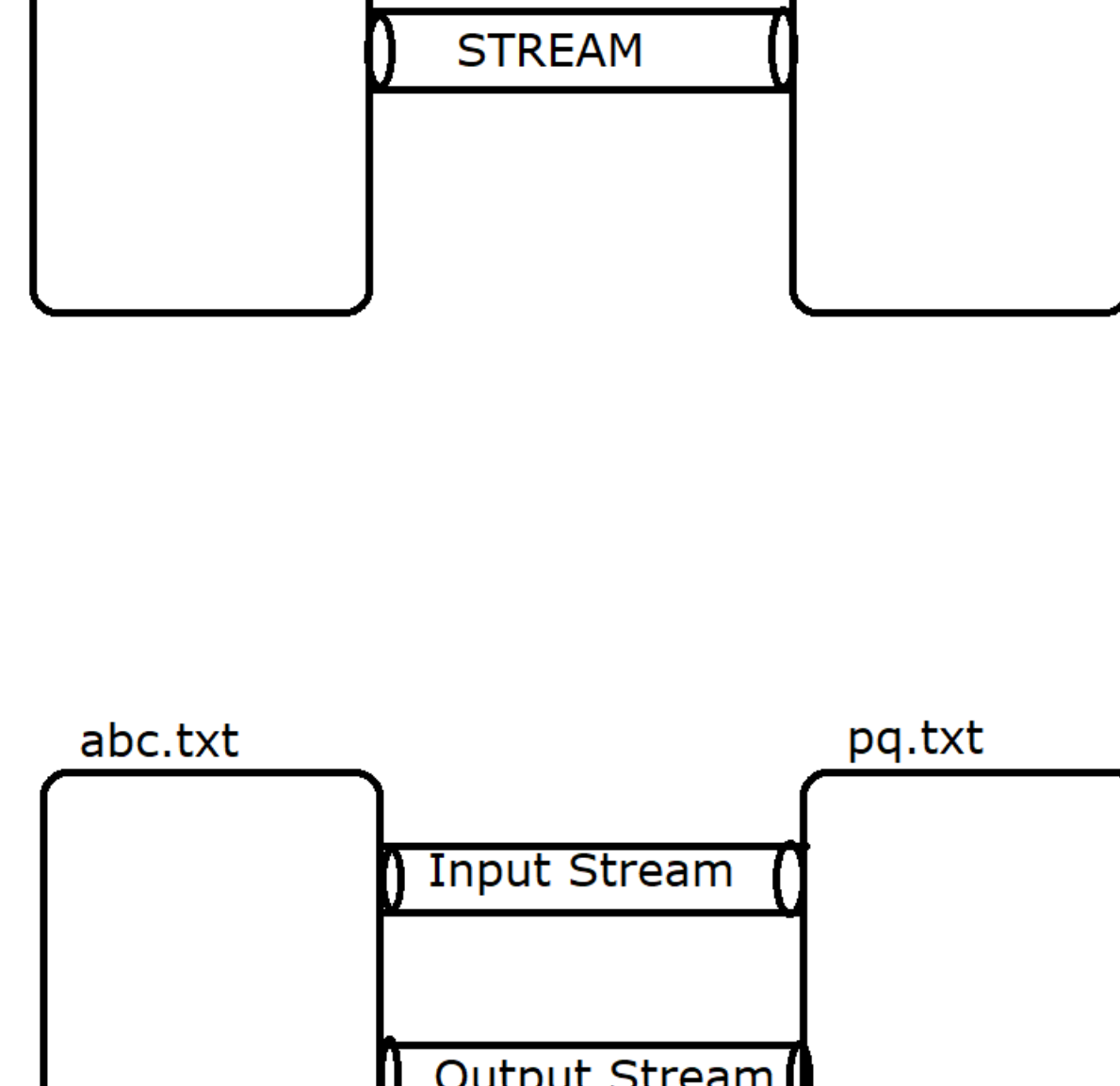
Once the execution of the program is completed, all the variables and fields values are destroyed from the memory (RAM). RAM is volatile memory that means we cannot use variable OR field value after program execution.

If we want to store the value of permennently so we can store these values in files (secondary memory) so in future we can retrieve the values from the files.

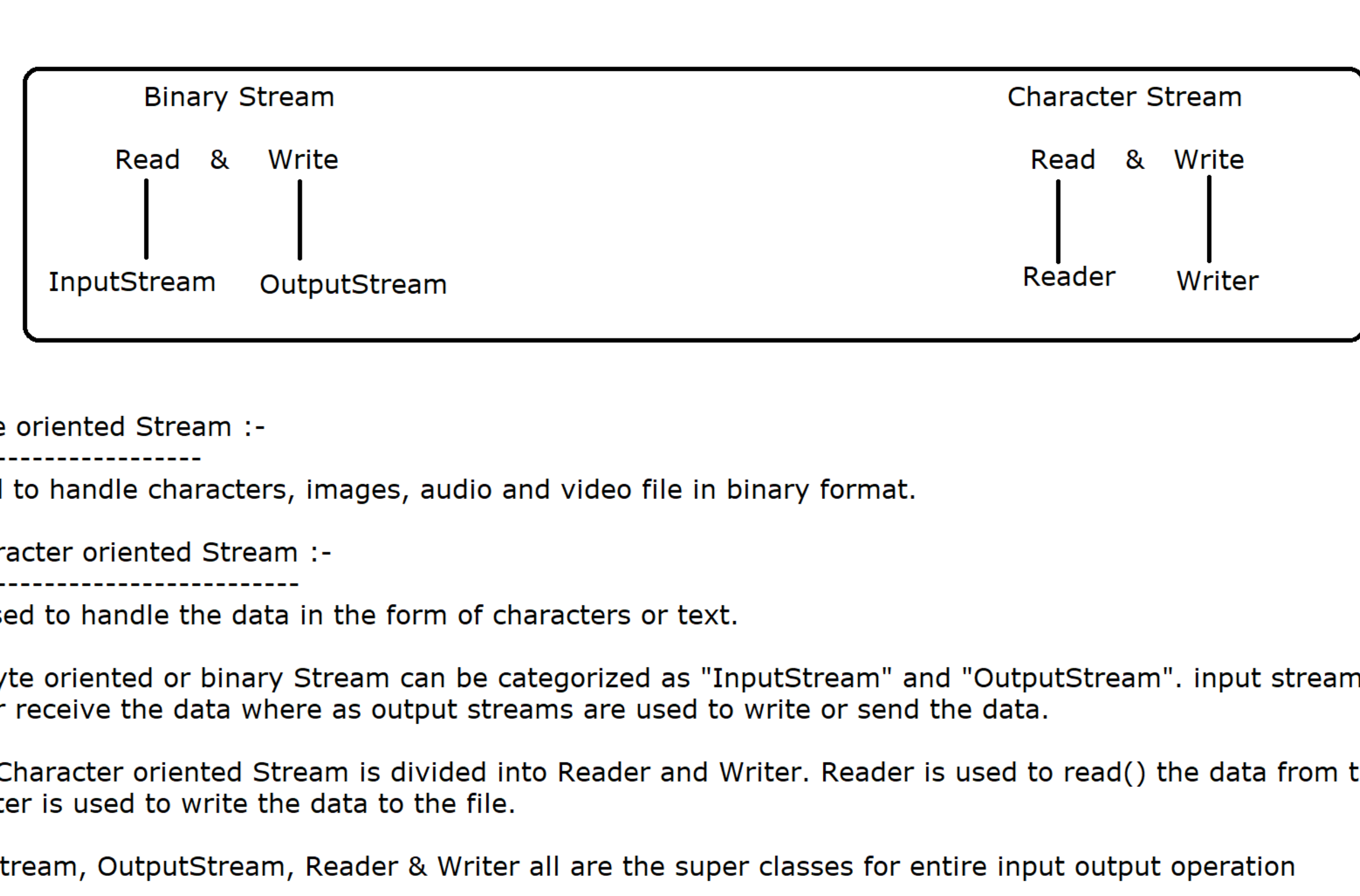
In order to perform read and write operation in the file we should use stream concept.

Streams in Java :

* A Stream is nothing but flow of data OR flow of characters to both the end of file system.



Types of Stream :



1) byte oriented Stream :-

It used to handle characters, images, audio and video file in binary format.

2) character oriented Stream :-

It is used to handle the data in the form of characters or text.

Now byte oriented or binary Stream can be categorized as "InputStream" and "OutputStream". Input streams are used to read or receive the data where as output streams are used to write or send the data.

Again Character oriented Stream is divided into Reader and Writer. Reader is used to read() the data from the file where as Writer is used to write the data to the file.

InputStream, OutputStream, Reader & Writer all are the super classes for entire input output operation

All Streams are represented by classes in java.io package.

FileOutputStream :

It is a predefined class available in java.io package. It comes under binary oriented Stream so, If we want to write the data to the file then the data must be available in binary format.

The main purpose of FileOutputStream class is, To create a new file (If file already exist then override/replace) as well as write the data in binary format in the specified file.

In order to write the data we can use a predefined method write(byte []b) which accepts byte array as a parameter.

How to convert the String text data into binary format :

String class has provided a predefined method called **getBytes()** through which we can convert the String (Collection of Characters) into byte array data as shown below :

```
String str = "Hello India";  
byte []b = str.getBytes();
```

//Program that describes how to convert String data into binary :

```
StringToBinary.java  
-----  
package com.ravi.binary_stream;  
  
public class StringToBinary {  
  
    public static void main(String[] args)  
    {  
        String str = "ABCDEF";  
  
        byte[] binaryData = str.getBytes();  
  
        for(byte data : binaryData)  
        {  
            System.out.println(data); //65 66 67 68 69 70  
        }  
    }  
}  
  
WAP to create a file and write some binary data to the file :  
-----  
package com.ravi.binary_stream;  
  
import java.io.FileOutputStream;  
import java.io.IOException;  
import java.io.PrintWriter;  
  
public class CreatingFileAndWritingTheData  
{  
    public static void main(String[] args) throws IOException  
    {  
        String filePath = "C:\\new\\Hyderabad.txt";  
  
        var fout = new FileOutputStream(filePath);  
  
        try(fout)  
        {  
            String data = "Hyderabad is a nice city, An IT City";  
            byte[] binaryData = data.getBytes();  
  
            fout.write(binaryData);  
        }  
        catch(IOException e)  
        {  
            System.err.println("IO exception encounter");  
        }  
        IO.println("Data stored");  
    }  
}
```

FileInputStream :

* It is a predefined class available java.io package
* It is used to read the binary data from the file.
* In order to read the data we can use a predefined method called read()

public int read() throws IOException

* It is a predefined non static method of FileInputStream class
* It is used to read a single character at a time and returns the UNICODE value of that particular character. [A -> 65]
* If there is no data in the file then it will return -1 which describes EOF (End of file)

ReadingTheData.java

```
package com.ravi.binary_stream;  
  
import java.io.FileInputStream;  
import java.io.IOException;  
  
public class ReadingTheData {  
  
    public static void main(String[] args) throws IOException  
    {  
        String filePath = "C:\\new\\Hyderabad.txt";  
        var fis = new FileInputStream(filePath);  
        try(fis)  
        {  
            while(true)  
            {  
                int data = fis.read();  
  
                if(data!=-1) //EOF  
                {  
                    break;  
                }  
                IO.print((char)data);  
            }  
        }  
        catch(IOException e)  
        {  
            e.printStackTrace();  
        }  
    }  
}
```

Working with Character Stream :

In order to perform read and write operation, We can use character Stream also.

Reader and Writer both are the abstract classes which represent super classes for perfroming all types of input and output operation using character stream.

FileWriter class :

It is a predefined class available in java.io package,By using this class we can create a file and write character data to the file.

By using this class we can directly write String (collection of characters) Or character array to the file.

Actually It is a character oriented Stream whereas if we work with FileOutputStream class, It is byte oriented Stream.

```
package com.ravi.character_stream;  
  
import java.io.FileWriter;  
import java.io.IOException;  
  
public class WritingData {  
  
    public static void main(String[] args) throws IOException  
    {  
        String filePath = "C:\\new\\India.txt";  
  
        var fout = new FileWriter(filePath);  
  
        try(fout)  
        {  
            String data = "India is a great country";  
  
            fout.write(data);  
        }  
        catch(IOException e)  
        {  
            e.printStackTrace();  
        }  
        IO.println("Data stored");  
    }  
}
```

FileReader :

It is a predefined class available in java.io package which will directly read the data in character format, here also we have read() method which will read the data character by character and returns UNICODE value of that character.

```
package com.ravi.character_stream;  
  
import java.io.FileNotFoundException;  
import java.io.FileReader;  
import java.io.IOException;  
  
public class ReadingData {  
  
    public static void main(String[] args) throws IOException  
    {  
        String filePath = "C:\\new\\India.txt";  
  
        var fr = new FileReader(filePath);  
  
        try(fr)  
        {  
            int data= 0;  
  
            while(true)  
            {  
                data = fr.read();  
  
                if(data==-1)  
                {  
                    break;  
                }  
                IO.print((char)data);  
            }  
        }  
        catch(IOException e)  
        {  
            e.printStackTrace();  
        }  
    }  
}
```